

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**4th Semester of B. Tech (CL) Examination****April - May 2018****ME 241 MATERIAL SCIENCE & TECHNOLOGY****Date: 07.05.2018, Monday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I**Q - 1 Answer Short Questions (Any Seven). [14]**

- 1) Draw the BCC, FCC, ECC and HCP crystal structure.
- 2) Define creep and fatigue of material.
- 3) Briefly explain Gibbs phase rule.
- 4) Write a difference between destructive and non-destructive testing.
- 5) Composite materials are a modern materials. Justify.
- 6) Why does corrosion occur in steel pipe connected to copper plumbing?
- 7) Differentiate between homogeneous and heterogeneous nucleation.
- 8) Define matrix phase and write functions of it.

Q – 2.a Explain the requirement of engineering materials. [03]**Q – 2.b Calculate the atomic packing factor of HCP structure. [05]**

Q – 2.c 1) Differentiate between edge dislocation and screw dislocation. [06]
 2) The external surface of specimen is considered as a defect. Justify.

Q – 2.d Draw the phase diagram of two metals which are completely soluble in liquid state and partially soluble in solid state. Draw the cooling curve and microstructure formation at single phase, dual phase, near eutectic and eutectic alloy. [07]**OR****Q – 2.d What is the significance of liquidus, solidus and solvus line in phase diagram? Explain lever rule. [07]****SECTION – II****Q – 3.a Draw Iron-Carbon diagram with all necessary details and state critical reactions of it. [07]****Q – 3.b State composition, specific properties and applications of low carbon steel. [06]****OR****Q – 3.b State composition, specific properties and applications of grey cast iron. [06]****Q – 4.a Explain any two methods for production of polymer matrix composites. Give applications of polymer composite materials. [07]****Q – 4.b Describe the mechanism of electrochemical corrosion by hydrogen evolution and oxygen adsorption. [06]**

Q – 5.a Explain liquid dye penetrant testing method. **[05]**

OR

Q – 5.a Explain sacrificial anode methods for corrosion prevention. **[05]**

Q – 5.b Explain the working of SEM with neat sketch. **[04]**